
APMTH 50 Problem Set 1

Due on Friday, February 17, by 5pm

Collaboration policy

Collaboration on homework and labs is encouraged, but each student must write solutions in their own words, and not copy them from anyone else. Please write the name(s) of your collaborator(s) at the top of the homework.

Submission guidelines

Please upload your homework as a single Word or pdf document. You do not need to include any code but you do include the relevant figures. For example your solution to problem 2 should include three separate figures, one figure for each of the cases in part b.

Problems

1. Competition: Predator–Prey

The following is a simple model to describe the interactions between predator and prey populations:

$$\begin{aligned}\frac{dR}{dt} &= \alpha R - \beta RF, \\ \frac{dF}{dt} &= -\gamma F + \delta RF,\end{aligned}$$

where $R(t) \geq 0$ and $F(t) \geq 0$ denote the rabbit (prey) and fox (predator) populations, and α, β, γ , and δ are positive parameters.

- (a) What are the units of the parameters α, β, γ , and δ ? Discuss the biological interpretations of these parameters.
- (b) Set $\alpha = 1.5, \beta = 0.5, \gamma = 0.4, \delta = 0.8$. Modify your code from lab and solve the system for initial populations of
 - i. 5 foxes and 0 rabbits,
 - ii. 5 foxes and 10 rabbits,
 - iii. 0 foxes and 10 rabbits.

Plot the rabbit and fox populations for each of these cases, and interpret the results.

2. Review of the original SIR Kermack–McKendrick paper

The original SIR paper by Kermack and McKendrick that was discussed in class is available on the course website. Please download the paper and answer the following questions. Note that we do not expect you to understand the paper in its entirety—one of the aims of this class is to glean the main points of a paper even if some of the details remain opaque.

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- (a) The paper makes a model of a flu epidemic in which the population is divided into three groups: the susceptible, the infected and the recovered. Explain what each of these groups are.
 - (b) A few years ago there was a global fear about a bird flu pandemic. Imagine there was a single person in Boston who had bird flu and nobody else was exposed to the disease. Explain what the infected, susceptible, and recovered populations would be in this case.
 - (c) The paper makes several assumptions about how a flu epidemic progresses through a population. List two of these assumptions.
 - (d) Explain what is plotted in the figure on page 714.
 - (e) The central result of the paper is that it proposes a formula for a threshold population density above which a flu epidemic will spread and potentially become large. What does this threshold population density depend upon? *[Hint: See the discussion on page 715. You can ignore or skim the dense mathematics that occurs in the pages before this—the main point is on this page.]*

3. Seasonal forcing

Exercise 8 of the lab assignment described a model of the common cold, whereby infected people are immediately returned to the susceptible category. Now consider a further generalization, whereby infected people first recover and receive temporary immunity, but over a longer time period that immunity is removed. This could be a reasonable model for flu, which slowly mutates, so the immunity gained from being infected will lose effectiveness over time. A suitable modification to the SIR equations is

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI + \alpha R, \\ \frac{dI}{dt} &= \beta SI - \gamma I, \\ \frac{dR}{dt} &= \gamma I - \alpha R,\end{aligned}$$

where the αR terms move people from the recovered category back to being susceptible.

- (a) For the common cold example in lab Exercise 8, the populations of S and I equilibrated at constant values that could be determined analytically. By doing a similar analysis here, find the equilibrium values of S , I , and R .
- (b) Suppose that α is very large. What will the approximate equilibrium values of S , I , and R be? Discuss how this relates to your answers from lab Exercise 8.
- (c) Numerically solve the equation for $S = 90$, $I = 10$, $R = 0$, $\beta = 0.02$, $\gamma = 1$, and $\alpha = 0.1$. Check that the steady-state values of S , I , and R match your solution from part (a).
- (d) **(Extra credit.)** Now consider a simple model of seasonal forcing, whereby β is no longer constant but oscillates according to the weather so that $\beta(t) = \beta_*(1 + 0.2 \cos \frac{2\pi t}{52})$,

where t is measured in weeks. Let $S = 99$, $I = 1$, $R = 0$ initially, and set $\gamma = 1$ and $\alpha = 0.04$. Solve the system numerically for three values of $\beta_* = 0.01, 0.0115, 0.013$ over the range $0 \leq t \leq 500$ and comment on the results.

4. High risk and low-risk populations

- (a) First, consider the standard SIR model,

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI, \\ \frac{dI}{dt} &= \beta SI - \gamma I, \\ \frac{dR}{dt} &= \gamma I.\end{aligned}$$

Suppose that $\beta = 0.1$, and $\gamma = 20$, and consider a population of one hundred people where initially $S = 90$, $I = 10$, and $R = 0$. Calculate the basic reproduction rate R_0 . Solve the differential equation system numerically for $0 \leq t \leq 1.2$, and by looking at the final steady-state value of R , estimate the total number of people who become sick.

- (b) Now suppose that the population in part (a) also comes into contact with a second, high-risk population. Let the number of susceptible, infected, and recovered people in this population be $S_H(t)$, $I_H(t)$, and $R_H(t)$ respectively. A combined model for this is given by the system of differential equations

$$\begin{aligned}\frac{dS}{dt} &= -S(\beta I + \beta_2 I_H), \\ \frac{dI}{dt} &= S(\beta I + \beta_2 I_H) - \gamma I, \\ \frac{dR}{dt} &= \gamma I, \\ \frac{dS_H}{dt} &= -S_H(\beta_2 I + \beta_3 I_H), \\ \frac{dI_H}{dt} &= S_H(\beta_2 I + \beta_3 I_H) - \gamma I_H, \\ \frac{dR_H}{dt} &= \gamma I_H.\end{aligned}$$

If $\beta_2 = 0$, then this system represents two independent SIR populations. However, if $\beta_2 > 0$ the two populations are coupled, and infected people in the high-risk population can infect people in the original population, and vice versa. Solve the model numerically for the parameter values of $S = 90$, $I = 10$, $R = 0$, $S_H = 100$, $I_H = 0$, $R_H = 0$, $\beta = 0.1$, $\beta_2 = 0.05$, $\beta_3 = 0.5$, and $\gamma = 20$ over the time interval $0 \leq t \leq 1.2$. Using the results, estimate the total number of people who become sick in both the original and high-risk populations.

- (c) Describe a real-life example where the situation considered in part (b) would be a good model. For your example, discuss how the model may influence public policy decisions, such as vaccination strategies.

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5. **(Extra credit.) A more realistic predator–prey model.** Consider again the predator–prey model from Exercise 1. In the absence of foxes, the model predicts that the rabbit population will grow without limit. In reality, one expects that the rabbit population will eventually plateau due to finite food resources available.

Consider a population of rabbits and foxes on an isolated island. Design a system of differential equations for the rabbit population $R(t)$, fox population $F(t)$, and also the total grass $G(t)$ on the island. Justify the terms in your equations. Solve the system numerically for two different initial conditions and comment on the results.

6. Estimate the number of hours it took to complete the homework, excluding the time spent on the extra credit portions.